

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

Ans:-

| No. | Common Hardware Issue      | Typical Symptom                                                           |
|-----|----------------------------|---------------------------------------------------------------------------|
| 1.  | Faulty RAM                 | PC does not boot, blank screen, or repeated beep sounds.                  |
| 2.  | Failed Power Supply (PSU)  | PC does not turn on or suddenly shuts down.                               |
| 3.  | Hard Drive/SSD Failure     | Slow boot, operating system errors, or drive not detected.                |
| 4.  | Overheating CPU            | PC becomes very hot, slows down, freezes, or shuts down unexpectedly.     |
| 5.  | Faulty Graphics Card (GPU) | No display, screen artifacts, or crashes during graphics-intensive tasks. |

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step Troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

Ans:-

### **Step 1 - Check the Wall Outlet**

- Plug another device (lamp/phone charger) into the same outlet.
- If the other device is also dead: the outlet, switch, or fuse is the issue. Try a different outlet.
- If the other device works: the outlet is OK, so move to the next step.

### **Step 2 - Inspect Charger Cable and Brick**

- Visually check the entire cable for cuts, kinks, and exposed wires.
- Check the brick for a burning smell, bulging, or loose parts.
- Ensure both ends (wall side and laptop side) are firmly plugged in.

### **Step 3 - Check the Laptop Charging Port**

- With the laptop OFF and unplugged, look into the DC-in port for dust, bent pins, or looseness.
- Gently wiggle the connector while plugged in (laptop off). If charging cuts in and out, the port is loose or damaged.

### **Step 4 - Check Adapter Rating**

- Compare the voltage (V), amperage (A), and wattage (W) on the charger label with the label on the laptop.
- If the charger is lower wattage or a different voltage, the laptop may power on but not charge.
- If the ratings match, move to the next step.

### **Step 5 - Try a Known-Good Charger**

- Connect a compatible charger from another person or the same laptop model.
- If the laptop charges: the original charger is faulty, so replace it.
- If it still does not charge: the fault is in the laptop, so move to the next step.

### **Step 6 - Perform a Power Reset**

- Shut down the laptop, unplug the charger, and remove the battery if it is removable.
- Hold the power button for 30 seconds to drain leftover charge.
- Plug in the charger without the battery, power on, then reinsert the battery.

### **Step 7 - Inspect the Battery**

- Check the battery contacts for dirt or corrosion and clean gently with a dry cloth.
- Look for swelling, such as a bulging case or a laptop that does not sit flat. If swollen, stop using it immediately, as it is a fire hazard.
- Run the laptop on the charger with the battery removed. If it works, the battery is dead and needs replacing.

### **Step 8 - Check for Overheating**

- Feel whether the laptop is very hot and make sure the vents are not blocked.
- Let it cool for 15 to 20 minutes, then try charging again, since many laptops stop charging when overheated.

### **Step 9 - Conclusion**

- Works with another charger: faulty adapter, so replace the charger.
- Works without the battery only: dead battery, so replace the battery.
- Port is loose or damaged: charging port fault, so get it repaired.
- Nothing works: possible motherboard or charging circuit fault, so take it to a technician.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

Ans: -

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## Tools Required

- Digital multimeter
- Standard USB phone charger (5V type)
- Wall outlet
- USB breakout connector (or a USB cable with the end cut open, only if your instructor allows)
- Optional: a virtual simulator such as Tinkercad Circuits

## Step 1 - Set Up the Multimeter

- Insert the black probe into the **COM** port and the red probe into the **VΩ** port.
- Turn the dial to **DC voltage (V=)**, range 20V.

## Step 2 - Connect the Charger

- Plug the charger into the wall outlet. Do not connect a phone.
- Use a USB breakout board, or expose the wires of a spare cable. In a standard USB cable, **red = +5V** and **black = ground (GND)**. The white and green wires are data lines, so do not touch them.
- Alternatively, touch the probes to the metal pins inside a USB-A port. The outer pins are +5V and GND, and the two middle pins are data. Be careful not to short the pins together.

## Step 3 - Measure

- Touch the red probe to the **+5V** pin or wire and the black probe to **GND**.
- Read the value on the display and record it.

## Step 4 - Compare with the Expected Range

- Expected range for USB: **4.75V to 5.25V**.
- If the reading is inside the range, the charger output is normal.
- If the reading is outside the range, the charger is faulty and should be replaced.

# Result Table

| Item               | Details                      |
|--------------------|------------------------------|
| Device tested      | USB phone charger (rated 5V) |
| Multimeter setting | DC voltage, 20V range        |
| Measured voltage   | 5.02 V DC                    |
| Expected range     | 4.75V to 5.25V               |
| Within range?      | Yes / No                     |

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?  
  
**Hint:** Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

Ans: -

### ATX 12V Rail Test – Step by Step

1. **Turn off the PC** – Shut down the PC and unplug it from the wall.
2. **Disconnect the ATX connector** – Disconnect the PSU's ATX connector from the motherboard.
3. **Set the multimeter** – Set it to **DC Voltage (V=)**.
4. **Identify wires** – Find the **yellow (+12V)** and **black (GND)** wires.
5. **Connect probes** – Place the **red probe on the yellow wire** and the **black probe on the black wire**.
6. **Turn on the PSU** – Power the PSU safely as required for the test setup.
7. **Read the voltage** – Check the voltage shown on the multimeter.
8. **Record the reading** – Example: **12.08V**.
9. **Compare the result** – Normal 12V rail should be around **12V; 11.5V or higher** is generally acceptable for this task.
10. **If below 11.5V** – It could indicate a **faulty or weak PSU**, which may cause random shutdowns.

**Measured Voltage: 12.08V**

**Result: Normal – PSU 12V rail is within the expected range.**